Effective Metric Response in MAAT Structural Cosmology

Gravitational Slip, Weyl-Potential Proxies, and Growth–Lensing Consistency

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Abstract

Previous MAAT cosmology papers promoted the structural projection observable from a diagnostic quantity to an effective source for linear matter growth. In Paper 43, this was implemented through a bounded projection kernel $\widehat{\mathcal{C}}_{\rm proj}(z)$ entering the growth coupling $\mu(z) = G_{\rm eff}/G = 1 + \eta_g \widehat{\mathcal{C}}_{\rm proj}(z)$. The resulting benchmark closed the first explicit growth-ODE layer, but it did not yet address metric perturbations, gravitational slip, or lensing-sensitive combinations of the scalar potentials.

This paper introduces a minimal effective metric-response benchmark. Working in the Newtonian-gauge scalar-potential language, we supplement the growth coupling by a gravitational-slip relation

$$\eta_{\rm slip}(z) = \frac{\Phi}{\Psi} = 1 + \beta_s \widehat{\mathcal{C}}_{\rm proj}(z),$$

and define the Weyl/lensing response

$$\Sigma(z) = \mu(z)\frac{1 + \eta_{\rm slip}(z)}{2}.$$

This extends the MAAT perturbative pipeline from growth response to metric response while remaining an effective, bounded, non-Boltzmann-code test.

Numerically, we scan $\eta_g \in [0, 0.08]$ and $\beta_s \in [-0.06, 0.06]$, giving 2501 metric-response branches. All scanned branches remain positive and stable in the tested sense. The compact growth-only comparison still prefers the ΛCDM limit $\eta_g = 0$, while β_s is unconstrained without lensing data. For a representative nonzero branch $(\eta_g, \beta_s) = (0.02, 0.03)$, the maximum deviations over $0 \leq z \leq 3$ are $|\mu - 1| \simeq 1.997\%$, $|\eta_{\rm slip} - 1| \simeq 2.995\%$, $|\Sigma - 1| \simeq 3.524\%$, and a Weyl-proxy deviation of $\simeq 2.731\%$.

The contribution is methodological: MAAT structural projection can be embedded into the standard metric-response language of linear cosmological perturbations without producing immediate pathologies. The result is not a weak-lensing likelihood, a CMB calculation, or evidence for modified gravity.

1 Purpose and Status

Paper 43 established the first explicit MAAT linear-growth equation by embedding the bounded projection kernel $\widehat{\mathcal{C}}_{\rm proj}(z)$ into the effective growth coupling $\mu(z)$. This was a necessary but incomplete perturbative step. Growth data probe matter clustering, but relativistic cosmology also probes the scalar metric potentials through lensing, integrated Sachs–Wolfe effects, and growth–lensing consistency.

The purpose of the present paper is therefore:

to extend MAAT structural cosmology from growth-only response to a minimal effective metric-response benchmark.

The status is deliberately conservative. This paper does not implement a Boltzmann solver, does not compute CMB anisotropies, does not include weak-lensing likelihoods, and does not claim a precision cosmological fit. It provides a reproducible bridge from the MAAT projection layer to the standard μ–Σ–slip language used in phenomenological tests of gravity.

2 Background: Scalar Metric Potentials

In Newtonian gauge, scalar perturbations of a spatially flat FLRW background can be written schematically as

$$ds^2 = -(1 + 2\Psi)\, dt^2 + a^2(t)(1 - 2\Phi)\delta_{ij} dx^i dx^j. \tag{1}$$

In minimally coupled general relativity with negligible anisotropic stress,

$$\Phi = \Psi. \tag{2}$$

Many effective modified-gravity parameterizations instead introduce two functions:

$$k^2\Psi = -4\pi G a^2 \mu(a)\rho_m \delta_m, \tag{3}$$

$$\Phi = \eta_{\rm slip}(a)\Psi. \tag{4}$$

The combination relevant for weak lensing is the Weyl potential $\Phi + \Psi$. It is conventional to define

$$k^2(\Phi + \Psi) = -8\pi G a^2 \Sigma(a)\rho_m \delta_m, \tag{5}$$

where

$$\Sigma(a) = \mu(a)\frac{1 + \eta_{\rm slip}(a)}{2}. \tag{6}$$

Thus Σ parameterizes the Weyl/lensing response channel: it measures how the lensing potential responds relative to the matter density contrast.

3 MAAT Metric-Response Ansatz

The bounded projection kernel is inherited from Paper 43:

$$\widehat{\mathcal{C}}_{\rm proj}(z) = \frac{\mathcal{C}_{\rm proj}(z) - \mathcal{C}_{\rm proj}(0)}{\mathcal{C}_{\rm proj}(z_{\rm max}) - \mathcal{C}_{\rm proj}(0)}, \qquad 0 \le \widehat{\mathcal{C}}_{\rm proj}(z) \le 1, \tag{7}$$

with $z_{\rm max} = 3$ in the present benchmark.

The growth response remains

$$\mu(z) = 1 + \eta_g \widehat{\mathcal{C}}_{\rm proj}(z), \tag{8}$$

where η_g controls the maximum fractional change of the effective Newtonian growth coupling over the tested interval.

The new metric-response ingredient is the slip ansatz

$$\eta_{\rm slip}(z) = \frac{\Phi}{\Psi} = 1 + \beta_s \widehat{\mathcal{C}}_{\rm proj}(z), \tag{9}$$

where β_s controls the maximum slip deformation. Operationally, a nonzero slip corresponds to an effective anisotropic-stress sector in the metric perturbations, even though no microscopic anisotropic stress model is derived here. Combining Eqs. (6)–(9) gives

$$\Sigma(z) = \left[1 + \eta_g \widehat{\mathcal{C}}_{\rm proj}(z)\right] \left[1 + \frac{\beta_s}{2}\widehat{\mathcal{C}}_{\rm proj}(z)\right]. \tag{10}$$

This construction is not interpreted as a fundamental variation of Newton's constant. It is an effective metric-response insertion of the same structural projection kernel already used in the growth sector.

4 Growth Equation

The matter growth equation is the same as in Paper 43:

$$\frac{d^2D}{dN^2} + \left[2 + \frac{d\ln H}{dN}\right]\frac{dD}{dN} - \frac{3}{2}\Omega_m(a)\mu(a)D = 0, \tag{11}$$

where $N = \ln a$ and $D(a)$ is normalized to $D(a = 1) = 1$. For $\eta_g = 0$, Eq. (11) reduces to the ΛCDM reference growth equation.

The growth observable is

$$f\sigma_8(z) = f(z)D(z)\sigma_{8,0}, \qquad f = \frac{d\ln D}{d\ln a}. \tag{12}$$

The compact $f\sigma_8$ comparison set used in Papers 40, 42, and 43 is reused only to quantify the growth channel. It does not constrain β_s, because β_s enters the metric/lensing sector rather than the growth equation. Equivalently, the source term of Eq. (11) depends on $\mu(z)$, not on $\eta_{\rm slip}(z)$; the slip channel becomes observable only through metric-potential probes such as lensing or ISW-type measurements.

5 Lensing and Consistency Proxies

Because this benchmark does not implement a weak-lensing likelihood, we report two diagnostic proxies rather than data-level observables. The first is a Weyl-amplitude proxy,

$$W_{\rm proxy}(z) = \Sigma(z)\frac{D_{\rm MAAT}(z)}{D_{\Lambda{\rm CDM}}(z)}. \tag{13}$$

The second is an E_G-like growth–lensing consistency proxy,

$$E_{G,{\rm proxy}}(z) = \Sigma(z)\frac{f_{\Lambda{\rm CDM}}(z)}{f_{\rm MAAT}(z)}. \tag{14}$$

These quantities are not replacements for lensing maps or galaxy–lensing cross-correlations. They are diagnostic projections showing how the metric-response layer would separate growth and lensing channels.

6 Numerical Setup

The benchmark uses the same Planck-like flat ΛCDM reference and projection parameters as Paper 43:

$$H_0 = 67.4\ {\rm km\,s^{-1}\,Mpc^{-1}}, \tag{15}$$
$$\Omega_{m0} = 0.315, \tag{16}$$
$$\Omega_{\Lambda 0} = 0.685, \tag{17}$$
$$\sigma_{8,0} = 0.811. \tag{18}$$

The scan ranges are

$$\eta_g \in [0, 0.08], \qquad \beta_s \in [-0.06, 0.06], \tag{19}$$

with 41 values of η_g and 61 values of β_s. This gives 2501 metric-response branches.

The representative nonzero branch used in the figures is

$$\eta_g = 0.02, \qquad \beta_s = 0.03. \tag{20}$$

The stability checks are deliberately minimal:

$$\mu(z) > 0, \qquad \eta_{\rm slip}(z) > 0, \qquad \Sigma(z) > 0, \qquad D(z) > 0, \qquad f(z) > 0. \tag{21}$$

7 Results

Table 1: Paper 45 metric-response scan summary.

Quantity	Value
η_g scan	$[0.00, 0.08]$ with 41 points
β_s scan	$[-0.06, 0.06]$ with 61 points
Total branches	2501
Stable / positive branches	2501/2501
Growth-only best η_g	0.0000
Growth-only β_s	unconstrained without lensing data
Best growth-only χ^2	12.2021

The growth-only compact comparison again prefers the ΛCDM limit $\eta_g = 0$. This is consistent with Paper 43. The slip parameter β_s is not constrained by the growth-only comparison, which is exactly why metric or lensing information is needed.

Table 2: Representative nonzero branch, $\eta_g = 0.02$, $\beta_s = 0.03$.

Quantity	Maximum absolute deviation for $0 \leq z \leq 3$
$\lvert\Delta D/D\rvert$	0.7656%
$\lvert\Delta f\sigma_8/f\sigma_8\rvert$	0.5441%
$\lvert\mu - 1\rvert$	1.9966%
$\lvert\eta_{\mathrm{slip}} - 1\rvert$	2.9949%
$\lvert\Sigma - 1\rvert$	3.5240%
Weyl-proxy deviation	2.7314%
E_G-proxy deviation	2.3019%

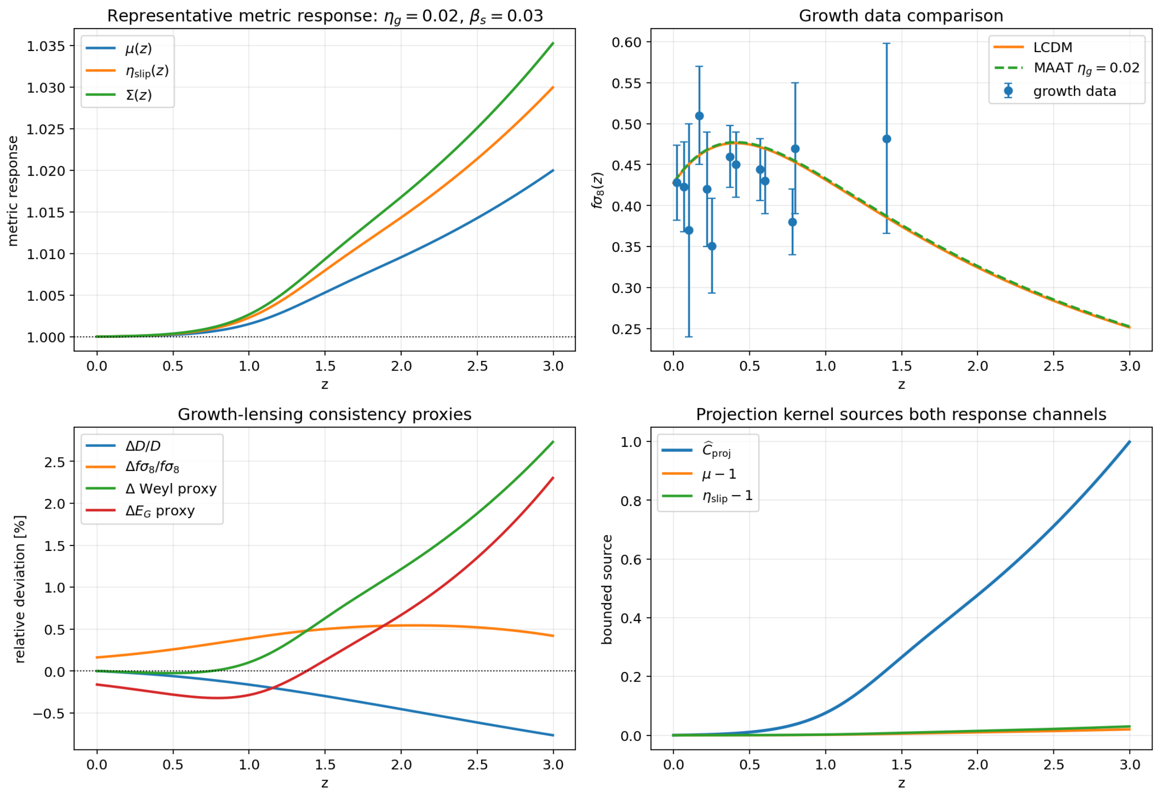

Figure 1: Paper 45 metric-response summary. Top left: representative metric response functions. Top right: compact $f\sigma_8$ comparison. Bottom left: growth, Weyl, and E_G proxy deviations. Bottom right: bounded projection kernel sourcing both response channels.

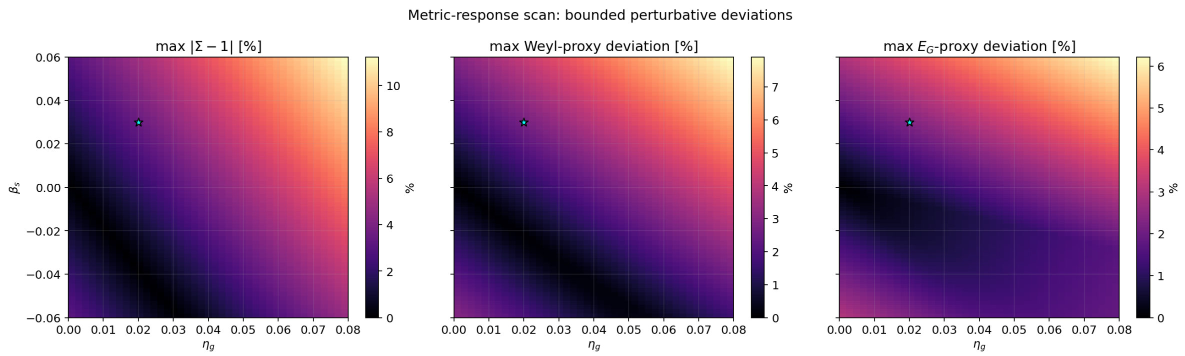

Figure 2: Metric-response scan over (η_g, β_s). The scan remains bounded and positive over the tested region. The representative branch is marked by a star.

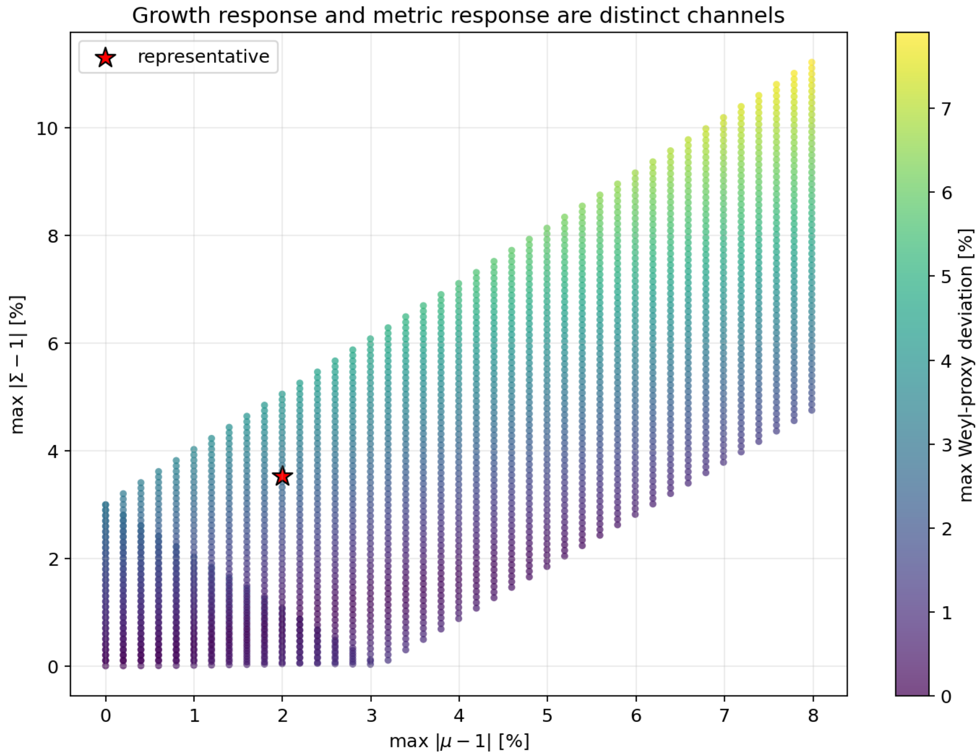

Figure 3: Growth response and metric response are distinct channels. At fixed growth coupling, changing β_s modifies Σ and the Weyl proxy without changing the growth-only χ^2.

8 Interpretation

Paper 45 adds the missing metric-response layer between growth-only perturbation dynamics and future lensing-level observables. The main conceptual result is:

> *MAAT projection structure can be coupled to matter growth and metric potentials as two distinct bounded response channels.*

This is important because growth data alone cannot determine whether a structural perturbation affects only matter clustering or also the Weyl potential. The parameter η_g controls the growth channel through $\mu(z)$. The parameter β_s controls the metric-slip channel through $\eta_{\mathrm{slip}}(z)$ and $\Sigma(z)$. This makes the two-channel structure explicit: η_g changes the effective source strength for matter perturbations, while β_s changes the relation between the two scalar potentials and therefore the Weyl/lensing response.

In the present compact comparison, $\eta_g = 0$ remains preferred by growth data, and β_s is unconstrained. Thus Paper 45 is not evidence for a nonzero metric response. Its contribution is to establish the formal bridge needed for future weak-lensing, ISW, and growth–lensing consistency tests.

9 Limitations

The limitations are substantial:

- no Boltzmann-code implementation,
- no CMB anisotropy calculation,
- no weak-lensing likelihood,

- no galaxy–lensing cross-correlation,
- no scale dependence in $\mu(k, z)$ or $\Sigma(k, z)$,
- no nonlinear structure formation,
- no MCMC inference or covariance treatment.

The benchmark should be read as a reproducible effective-theory bridge, not as a data-level cosmological constraint.

10 Reproducibility

The code and outputs are stored in:

`experiments/maat_metric_response_paper45/`

Run:

```
python3 maat_metric_response_solver_v01.py
```

The script writes:

- `outputs_paper45/paper45_summary.json`,
- `outputs_paper45/paper45_metric_curves.csv`,
- `outputs_paper45/paper45_metric_scan.csv`,
- `outputs_paper45/paper45_growth_metric_comparison.csv`,
- three PNG figures used in this paper.

The public repository is:

https://github.com/Chris4081/structural-selection-principle

Data attribution and license note. The Planck-normalised reference parameters and compact $f\sigma_8$ comparison points are external scientific data and should be cited to the original publications/collaborations when reused or discussed. The CSV tables and figures in this repository are derived reproducibility artifacts only. No endorsement by the Planck Collaboration, survey collaborations, or original data authors is implied.

11 Conclusion

We introduced a minimal effective metric-response benchmark for MAAT structural cosmology. The construction extends the Paper-43 growth response $\mu(z) = 1 + \eta_g \widehat{\mathcal{C}}_{\rm proj}(z)$ by adding gravitational slip $\eta_{\rm slip}(z) = 1 + \beta_s \widehat{\mathcal{C}}_{\rm proj}(z)$ and the Weyl/lensing response $\Sigma(z) = \mu(1 + \eta_{\rm slip})/2$.

All scanned branches remain positive and stable in the tested interval. The compact growth comparison still prefers $\eta_g = 0$ and cannot constrain β_s. For a representative nonzero branch, the metric response remains at the few-percent level.

The next hard step is not another proxy, but a full perturbation implementation with scale dependence and weak-lensing/CMB likelihoods.

References

[1] C.-P. Ma and E. Bertschinger, "Cosmological Perturbation Theory in the Synchronous and Conformal Newtonian Gauges," *Astrophys. J.* **455**, 7 (1995).

[2] L. Amendola, M. Kunz, and D. Sapone, "Measuring the dark side with weak lensing," *JCAP* **04**, 013 (2008).

[3] A. Joyce, B. Jain, J. Khoury, and M. Trodden, "Beyond the Cosmological Standard Model," *Phys. Rep.* **568**, 1–98 (2015).

[4] T. Clifton, P. G. Ferreira, A. Padilla, and C. Skordis, "Modified Gravity and Cosmology," *Phys. Rep.* **513**, 1–189 (2012).

[5] Planck Collaboration, "Planck 2018 results. VI. Cosmological parameters," *Astron. Astrophys.* **641**, A6 (2020).

[6] S. Alam et al., "The clustering of galaxies in the completed SDSS-III Baryon Oscillation Spectroscopic Survey: cosmological analysis of the DR12 galaxy sample," *Mon. Not. R. Astron. Soc.* **470**, 2617–2652 (2017).

[7] C. Krieg, "Linear Perturbations and Structure Growth in Structural-Selection Cosmology," preprint (2026).